

1 Reading papers

Two sources are named on the Read shelf of the site, and they answer different questions. Keshav gives a procedure for deciding how much of a paper to read. Eisner gives a posture for reading it once that decision is made. A third method is already in use in this project, the twenty five question scheme behind `Literature/reading_insights.md`, and it does something neither of the other two does.

1.1 Keshav: three passes, and the decision to stop

The method exists to prevent one failure, stated in the abstract: reading is “rarely taught, leading to much wasted effort.” The structure is a sequence of passes rather than a single read, and each pass ends in a decision about whether to continue.

The first pass is “a quick scan to get a bird’s-eye view of the paper” and “should take about five to ten minutes.” Five steps: read the title, abstract, and introduction carefully; read the section and subsection headings and ignore everything else; glance at the mathematical content to determine the theoretical foundations; read the conclusions; glance over the references, ticking off the ones already read.

At the end of it the reader answers the five Cs, quoted in full:

1. **Category.** “What type of paper is this? A measurement paper? An analysis of an existing system? A description of a research prototype?”
2. **Context.** “Which other papers is it related to? Which theoretical bases were used to analyze the problem?”
3. **Correctness.** “Do the assumptions appear to be valid?”
4. **Contributions.** “What are the paper’s main contributions?”
5. **Clarity.** “Is the paper well written?”

The pass ends in a stopping decision, and the paper names three legitimate reasons to stop: the paper does not interest you, you do not know enough about the area to understand it, or the authors make invalid assumptions. One pass is called “adequate for papers that aren’t in your research area, but may someday prove relevant.”

The second pass reads “with greater care, but ignore details such as proofs,” and “should take up to an hour for an experienced reader.” Two instructions carry it. Look carefully at the figures: “Are the axes properly labeled? Are results shown with error bars, so that conclusions are statistically significant? Common mistakes like these will separate rushed, shoddy work from the truly excellent.” And mark relevant unread references for further reading. The paper quotes a suggestion from Dominik Grusemann to “note down terms you didn’t understand, or questions you may want to ask the author,” and observes that for a referee those notes become the review and the material to defend it with at the committee meeting.

The test of the second pass is stated as an ability rather than a feeling: “You should be able to summarize the main thrust of the paper, with supporting evidence, to someone else.”

When the second pass fails, the paper names the causes and the three options. Causes: unfamiliar subject matter and terminology, a proof or experimental technique the reader does not know, poor writing with unsubstantiated assertions and forward references, or the hour. Options: set it aside, return after reading background material, or persevere into the third pass.

The third pass has one governing idea. “The key to the third pass is to attempt to virtually re-implement the paper: that is, making the same assumptions as the authors, re-create the work. By comparing this re-creation with the actual paper, you can easily identify not only a paper’s innovations, but also its hidden failings and assumptions.”

The instruction that follows is the one that matters for review work: “You should identify and challenge every assumption in every statement.” The pass takes “many hours for beginners and more than an hour or two even for an experienced reader,” and ends when the reader can reconstruct the structure of the paper from memory.

1.2 Keshav: the literature survey, in three steps

The same paper gives a procedure for entering an unfamiliar area, and it is a concrete alternative to searching by keyword.

Step one. Find five recent highly cited papers in the area, do one pass on each, then read their related work sections. If that yields a recent survey, “Read the survey, congratulating yourself on your good luck.”

Step two. “Find shared citations and repeated author names in the bibliography. These are the key papers and researchers in that area.” Download those papers. Then look at where those researchers have published recently, which identifies the top venues in the field.

Step three. Go to the recent proceedings of those venues and scan for related work. Those papers plus the ones set aside are the first version of the survey. Make two passes through them, and if they all cite a paper not yet found, obtain it and iterate.

The procedure is worth noting for what it does not use. It does not start from a query. It starts from a small seed set and follows citation structure outward, which is the same operation as the bidirectional citation chasing in the literature review procedure.

1.3 Eisner: how to read once the decision is made

Eisner’s advice is about posture, and four instructions are usable directly.

Skim first, “skipping over anything that would take much mental effort,” to establish direction and novelty, then return and read carefully. The order is the same as Keshav’s, arrived at independently.

Write while reading. “Write as you read. This keeps your attention focused and makes you engage with the paper.” Two levels of note are recommended: low level annotations on specific content, and a high level summary for future retrieval.

Read creatively. Identify the good ideas, consider applications the authors did not, and ask what you would do next. Question what is unclear, fill in missing steps, and connect the paper to other work.

When stuck, change material rather than pushing harder. “When the going gets tough, switch to

background reading,” meaning textbooks, tutorials, or review articles rather than more of the same dense original.

On selective reading, Eisner permits skimming a technical section when it explains familiar material, is overly specialized, or is too advanced, provided the reader still grasps what the method accomplishes.

1.4 The scheme already in use here

The twenty five question scheme behind `reading_insights.md` is a third kind of reading, and it should not be confused with the other two. Keshav and Eisner describe reading a paper. The scheme describes reading a paper for a purpose that is fixed in advance.

Its questions are not about the paper’s own argument. They are about what the paper supplies to a submission in progress: the Heilmeier framing, the single-sentence story, which contribution leads, who cares, reference template papers, section structure and page allocation, the twelve page section budget, threat model prominence, venue conventions, most cuttable content, failure modes, artifact sufficiency, where to publish, and so on. Every insight is recorded in the form *From <source>: ...*, one bullet per question per source, so that a later reader can collect one question across seventeen sources and see the spread of answers.

Three properties of that design are worth naming, because they are what make it work and they are reusable.

The question set is fixed before reading starts. The same twenty five questions are asked of every source. That is what makes the answers comparable across sources, and it is what allows a cross-source consistency pass to find contradictions.

Contradictions are recorded rather than resolved. The file’s own editorial section lists five framing contradictions across the seventeen sources and states that they “are EDITORIAL decisions the writer must make; they cannot be resolved by patching individual bullets.” A reading system that averages its sources loses this.

Quality is audited separately from insight. The `quality_*.md` files score each bullet for depth and check it against the source, which is a separate operation from generating the bullet. Self review by the same pass does not substitute for it.

1.5 What the measurement literature adds

Three findings change how much to trust a read, and all three concern the gap between reading an abstract and reading a paper.

Abstracts decide inclusion badly. Across 139,467 citations and 329,332 screening decisions by 86 reviewers, the total error rate for abstract screening was 10.76 percent. Of the citations screened, 18.07 percent passed the abstract stage and 5.48 percent survived to final inclusion. About two thirds of what an abstract admits is overturned once the body is read. A paper included on the strength of its abstract has not been read.

Extraction is accurate until the information is absent. A study of structured extraction reported 92.4 percent accuracy overall with a 95 percent confidence interval of 89.5 to 94.5, and false data produced in 5.2 percent of cases. The number that matters is the conditional one: reproducibility fell to 77.2 percent when the information was not reported in the paper, and the

failures took the form of plausible but incorrect content. The lesson is that a question a paper does not answer is where fabrication enters, so *not stated* has to be an available answer.

An operating rule from a review of 172 studies. Mean extraction precision was 83.0 percent and recall 86.0 percent, and the policy the authors adopted is worth copying without change: for categories with precision below 80 percent, assign a manual reviewer to validate and correct. Performance metrics were among the lower accuracy categories, which is the category most likely to be reused.

1.6 Putting the three together

The three sources answer three different questions, and a reading session uses all of them in order.

Keshav decides how far to read, and gives a stopping rule at each depth. Eisner governs what happens during the read: write while reading, question actively, switch material when stuck. The fixed question scheme decides what to extract, and keeps the extraction comparable across sources so contradictions surface instead of averaging out.

The measurement literature adds one constraint that overrides all three. A claim sourced from an abstract is marked as such and is not treated as a claim about the paper. Where a question has no answer in the paper, the answer recorded is that the paper does not address it.

1.7 Sources

- S. Keshav, *How to Read a Paper*, ACM SIGCOMM Computer Communication Review 37(3), 2007. Quotations are from the version of 17 February 2016, read in full, text extracted locally from the file at `svr-sk818-web.cl.cam.ac.uk/keshav/papers/07/paper-reading.pdf` after a fetch summary failed to return the text. Sections 2.1 to 2.3 and 3.
- Jason Eisner, *How to Read a Technical Paper*, 2009, `cs.jhu.edu/~jason/advice/how-to-read-a-paper.html`. Read as a web page. Quotations are short and are marked where they are exact.
- The twenty five question scheme and its insight format are this project's own, in `Literature/reading_insights.` and the accompanying `PaperReads/quality_*.md` files.
- The abstract screening figures are from a study of error rates in abstract screening across 86 reviewers. The extraction figures are from a study of structured extraction and from a review of 172 studies. Both reach this document through the research agents and literature review note, which records the full citations and marks which were read in full.

The comparison in the final subsection is not in any of the sources. It sets three independent methods against each other, which none of the three authors does.